

Product-market fit for UK energy operators

UK energy operators (operators of essential services) procure OT security against the Ofgem CAF Overlay, the NIS Regulations and IEC 62443, embedding audit rights and incident-response expectations into contracts. The table below sets out what they require when buying, and how OTShield meets each requirement, including the new AI detection layer developed in this project.

Table 1. Utility provider requirements mapped to OTShield

Utility requirement	How OTShield meets it
High-confidence detection, very low false positives	Deception: any decoy interaction is hostile by construction. New AI/ML anomaly detection reduces missed and noisy alerts further.
No disruption to real-time control or live controllers	Decoys are isolated; legitimate traffic never touches them; passive DPI only. Zero risk to live plant.
Detect AND respond, not just observe	Active defence plus incident-response workflow and SIEM/SOAR integration, beyond visibility-only tooling.
Coexist with existing tooling (no rip-and-replace)	Coexists with Dragos / Claroty / Nozomi, adding the deception, AI detection and copilot layers they lack.
Support for unpatchable legacy systems	Auto-discovers and risk-scores legacy assets; shows how an attacker would reach and abuse them.
On-premises, air-gapped, no data leaves the plant	Runs fully offline; local models; signed offline update bundles keep it current without internet.
Reduce analyst workload (skills scarcity)	Offline LLM-assisted copilot interprets and prioritises detections and makes safe threat hunting practical.
Alignment with NIS, NIS2, IEC 62443, NCSC CAF	Designed and evidenced against these frameworks; auditable findings ledger and signed SBOM.

Competitive comparison

The nearest available solutions are asset-visibility and detection platforms; IT-centric deception tools are a partial alternative. OTShield is the only option combining energy-native deception, a new AI detection layer and an offline analyst copilot in one platform that coexists with the incumbents.

Table 2. Capability comparison

Capability	OTShield	Dragos / Claroty / Nozomi	IT-centric deception	Generic IT IDS
Energy-native protocol-aware decoys	Yes	No	Partial	No
Active defence (detect AND respond)	Yes	Partial	Partial	No
AI/ML anomaly detection on OT traffic	Yes	Partial	No	Partial
AI attacker-behaviour classification	Yes	Partial	No	No
Offline analyst copilot (local LLM)	Yes	No	No	No
Very low false positives	Yes	Partial	Yes	No
No risk to live controllers	Yes	Yes	Yes	Partial
Air-gapped / fully offline	Yes	Partial	Partial	Partial
Coexists with incumbents	Yes	No	Partial	Yes
NIS / IEC 62443 / NCSC CAF aligned	Yes	Yes	Partial	Partial

Yes

Partial

No

Ratings reflect typical product positioning; incumbent strengths in asset visibility are acknowledged.

Market fit, segments and pricing

Target segments (energy first, then transferable)

Primary - UK energy operators

Generation, transmission, distribution, downstream gas and electricity within NIS scope. The project's host operator is the reference deployment.

Adjacent CNI

Water and transport operators with the same OT detection gap; the energy proof point transfers directly.

Defence and government

MOD, NCSC, NPSA and defence primes running ICS protection and supply-chain assurance.

ICS vendor red teams and allies

Siemens / Rockwell / Schneider-class internal programmes; Five Eyes and NATO-aligned markets in time.

Commercial pricing (published tiers)

STARTER GBP 20,000 / yr Pilot, real attacker detection. Up to 50 assets.	GROWTH GBP 50,000 / yr Visibility to active defence. Up to 500 assets. NIS2 reporting.	ADVANCED GBP 75,000 / yr Full-scale CNI detection & response. Unlimited assets.
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Product-market fit in one line

OTShield is the only platform that combines energy-native deception, a new AI threat-detection layer and an offline analyst copilot, meeting the exact requirements UK energy operators buy against (low false positives, no operational risk, detect-and-respond, coexistence, legacy support, air-gapped, NIS/IEC 62443/CAF aligned) where the incumbents stop at visibility and alerting. This project proves and qualifies that fit inside a real energy operator.